Optimization of Electrical and Instrumentation Deliverable Extraction from Piping and Instrumentation Diagrams via Hybrid Deterministic-Generative Artificial Intelligence and Labeled Property Graphs

Abstract

The engineering, procurement, and construction (EPC) sector is currently navigating a profound digital transformation, yet the extraction of critical Electrical and Instrumentation (E&I) deliverables from Piping and Instrumentation Diagrams (P&IDs) remains a persistent, high-cost manual bottleneck. This research investigates the "Contextual Integrity Gap," defined as the catastrophic failure of traditional, uncontextualized machine learning models to synthesize the functional engineering logic required for safe industrial deployment. While contemporary object detection models like YOLOv8 and advanced Optical Character Recognition (OCR) engines achieve significant spatial accuracy, they lack the inherent capability to respect non-negotiable regulatory standards such as ANSI/ISA-5.1-2024 and IEC 61511:2016. This study proposes a novel five-layer hybrid architecture that integrates probabilistic visual perception with deterministic engineering guardrails. The architecture utilizes YOLOv8 for symbol localization, a multi-stage OCR pipeline for tag extraction, and a topological layer centered on Labeled Property Graphs (LPG) to reconstruct plant connectivity. Central to this system is a cognitive layer leveraging Graph-based Retrieval-Augmented Generation (Graph-RAG), which mitigates large language model (LLM) hallucinations by grounding generative reasoning in validated graph-relational data. Evaluation of the prototype demonstrates 100% compliance with tagging conventions and zero-violation adherence to functional safety interlock logic. This methodology establishes a new benchmark for standards-aligned artificial intelligence in heavy industry, transitioning legacy documentation from static images into high-fidelity, queryable digital assets.

Introduction: The Industrial Imperative for Semantic Digitization

Within the highly complex, capital-intensive, and heavily regulated process industries—a domain encompassing oil and gas refineries, petrochemical manufacturing plants, nuclear power generation facilities, and large-scale municipal water treatment utilities—Piping and Instrumentation Diagrams (P&IDs) represent the absolute foundational blueprints for all engineering operations. These intricate, high-density schematic drawings serve as the primary reference documents that dictate the physical equipment layout, the overarching process control philosophy, the thermodynamic fluid flow connectivity, and the critical safety interlock systems required to maintain stable plant operations. A P&ID acts as a visual manifestation of the plant's operational logic, illustrating the functional relationship between massive mechanical equipment and the delicate instrumentation and control loops that govern their behavior.

From these two-dimensional visual representations, Electrical and Instrumentation (E&I) engineers are tasked with the massive responsibility of manually translating graphical schematics into an array of critical, downstream structured engineering deliverables. The most vital of these structured documents include the Instrument Index, the Input/Output (I/O) List, the Alarm List, and the automated Trip List, frequently formalized as a Cause and Effect Matrix. Historically, the preparation of these deliverables has been an entirely manual process strictly dependent on the subjective visual interpretation of domain experts. The scale of modern industrial engineering projects dramatically exacerbates this challenge; a single brownfield expansion or greenfield construction project typically encompasses thousands of individual P&ID sheets. Each sheet routinely contains hundreds of discrete symbols representing isolation valves, centrifugal pumps, pressurized vessels, and specialized measurement transmitters, all interconnected by a labyrinth of primary process piping and secondary electrical signal lines.

The financial burden of this manual process is staggering; industrial analyses indicate that manual instrument data extraction and cross-checking can cost a single engineering office upwards of $1.5 million annually just in expended labor hours. However, the consequences of manual transcription errors in this domain extend far beyond mere administrative inefficiencies; they represent severe, latent threats to industrial safety. The misclassification of critical signals—such as erroneously logging a discrete digital input (DI) as an analog input (AI) in the E&I database—can completely derail the physical wiring and software programming of the Basic Process Control System (BPCS). Even more critically, the incorrect mapping of safety shutdown logic, or the erroneous transcription of an alarm setpoint limit from the P&ID to the control system database, can fatally compromise the facility's Safety Instrumented System (SIS).

A seemingly negligible human oversight—such as missing a High-High pressure switch (PSHH) interlock on a distillation column—can silently invalidate a facility's rigorous Layer of Protection Analysis (LOPA). During an actual overpressurization event, this documentation failure could precipitate catastrophic safety hazards, cause unexpected environmental chemical releases, or result in massive financial losses due to unplanned operational downtime. This systemic failure to bridge raw visual data with engineering intent is formally defined as the "Contextual Integrity Gap". Current artificial intelligence models lack deep, deterministic integration with established, non-negotiable engineering standards, such as the International Society of Automation’s ISA-5.1 standard for instrument tagging and the International Electrotechnical Commission’s IEC 61511 standard for functional safety logic.

The primary objective of this research is to bridge this gap by engineering a standards-aligned, hybrid artificial intelligence pipeline capable of automatically extracting,

interpreting, and generating critical E&I deliverables directly from unstructured P&IDs. This study posits that true semantic understanding of engineering diagrams cannot be achieved through probabilistic computer vision alone. Instead, visual perception must be securely layered beneath deterministic engineering guardrails and evaluated using highly constrained generative AI. By integrating deep learning for visual localization with a Graph-Retrieved Augmented Generation (Graph-RAG) framework, this system enforces rigorous adherence to industrial doctrine while eliminating the risk of generative hallucinations common in unconstrained large language models.

Literature Review: The Technical Evolution of P&ID Digitization

The academic and industrial pursuit of automated P&ID digitization has been the subject of intense, multidisciplinary research over the past decade. A comprehensive review of the existing literature reveals that the methodological approaches to solving this problem have evolved through four distinct technical stages, each attempting to overcome the limitations of its predecessor in capturing the high density and entanglement of information inherent in complex engineering drawings (EDs).

Stage 1: Classical Image Processing and Morphological Rule-Based Systems

The foundational era of engineering drawing digitization relied heavily on deterministic image processing, morphological mathematics, and hard-coded heuristic rules. These early computational frameworks typically utilized sequential preprocessing steps to clean noisy, degraded, and scanned legacy diagrams. Techniques such as global binarization, adaptive local thresholding (Otsu's method), and noise filtering were employed to mathematically separate the foreground drafting ink from background artifacts. This was often followed by skeletonization or morphological thinning—a mathematical transformation designed to iteratively reduce the pixel volume of thick process lines to a single-pixel width.

Theoretically, this operation allowed algorithms to trace continuous line connectivity, map pipe routing, and construct rudimentary "netlists". For symbol recognition, classical approaches employed strict contour analysis combined with manually engineered visual descriptors. Pioneering research by Tan et al. (2016) demonstrated a specialized framework utilizing Local Binary Patterns (LBP) as a primary visual descriptor, capturing foundational shape information regardless of ambient noise by comparing intensity of neighboring pixels. To achieve scale stability, Tan et al. (2016) integrated Spatial Pyramid Matching (SPM), allowing the model to compute local blocks of a sub-image rather than the global geometry. This system circumvents the requirement for massive training datasets, functioning with only a single template example per component type.

However, the literature consistently documents extreme algorithmic fragility in rule-based pipelines. As Tan et al. (2016) and Kang et al. (2019) explicitly noted, a slight shift

in drafting style or a rotation of a symbol renders template-matching algorithms obsolete, necessitating continuous manual recalibration. Kang et al. (2019) attempted to overcome these limits by applying sliding-window blob analysis to track lines as continuous masses, distinguishing between primary process lines and thin utility lines via aspect ratios. Despite achieving a reported 16-fold reduction in manual drafting hours, morphological thinning operations frequently destroyed vital semantic information, such as flow directionality arrows, ruining downstream logic extraction required for safety modeling.

Stage 2: Deep Learning and the Shift to Uncontextualized Object Detection

The democratization of deep Convolutional Neural Networks (CNNs) initiated a paradigm shift, reframing digitization as a multi-class object detection problem. Researchers pivoted toward robust, automatic feature learning capabilities, improving spatial localization across diverse standards. Single-stage detectors like YOLOv3 were successfully applied to localize dozens of symbol classes simultaneously across high-resolution diagrams. Elyan et al. (2020) customized the YOLO architecture to predict bounding boxes for 25 distinct symbol classes, achieving recall rates exceeding 94%.

Despite bounding-box successes, Stage 2 exposed a severe "class imbalance" problem. Standard components like manual gate valves appear thousands of times in training sets, while critical specialized instruments appear rarely, causing neural networks to bias toward majority classes. To combat this, Elyan et al. (2020) integrated Deep Generative Adversarial Networks (GANs), specifically the Multiple Fake Class GAN (MFC-GAN), to synthesize realistic artificial variations of minority-class symbols. This innovation allowed models to learn from as few as two real-world instances, elevating detection rates for rare equipment to viable levels.

Parallel to single-stage detection, two-stage frameworks like Faster R-CNN with ResNet-50 backbones were explored to maximize precision on small, tightly clustered components. Gao et al. (2020) proposed a "feature grouping strategy" to detect components based on similar aspect ratios, establishing a reliable spatial baseline for nuclear fault tree analysis. Furthermore, Yun et al. (2020) proposed suppressing false positives triggered by graphical noise (revision clouds, grid lines) by utilizing Deep Adaptive Clustering (DAC) to group "dummy" instances into negative classes, achieving a dummy detection rate of only 0.39%. However, the fundamental limitation of Stage 2 remained: the models yielded only isolated spatial coordinates without functional context.

Stage 3: Topological Graph-Based Modeling and LSTM-OCR Integration

Subsequent research acknowledged that intelligent digitization required reconstructing the diagram's physical topology alongside robust extraction of alphanumeric tagging data. In this paradigm, equipment and instrument bubbles are treated as vertices

(nodes) in a directed graph, while pipelines and signal wires act as edges mapping fluid and signal flow. Gao et al. (2020) developed dictionaries representing component-to-pipe relationships via geometric intersection rules, generating structural representations required for flow analysis.

Extracting textual annotations—the foundation of Instrument Indexes and I/O Lists—posed challenges as standard OCR engines perform poorly on engineering documents due to vertical stacking and text-to-line overlap. To solve this, Jamieson et al. (2020) deployed the Efficient and Accurate Scene Text (EAST) detector, a fully convolutional network (FCN) capable of predicting rotational angles for words amidst graphical clutter. Crops from EAST were fed into Tesseract v4, utilizing Long Short-Term Memory (LSTM) sequence recognizers to decode character strings. While Jamieson et al. (2020) achieved an 86% recognition rate, accurate association of text strings to symbols remained unreliable. The majority of literature relied on "closest-distance" or Euclidean proximity heuristics, which fail in dense schematics where a tag might physically sit closer to an unrelated pump or cooling line. This Euclidean failure mode introduces unacceptable data integrity risks for safety-critical documentation.

Stage 4: Emerging Paradigms in Vision-Language Models and Graph-RAG

The current frontier in P&ID research formally acknowledges that purely visual algorithms and isolated OCR engines cannot independently overcome the Contextual Integrity Gap. Modern data extraction requires reasoning over thermodynamic and electrical relationships. Consequently, contemporary literature has introduced querying and reasoning over P&IDs by integrating Knowledge Graphs (KGs) with Large Language Models (LLMs).

Gupta et al. (2025) addressed the gap between raw documentation and natural language retrieval by introducing the PIDQA dataset, a compilation of 64,000 question-answer pairs derived from 500 P&ID sheets. Queries were categorized to benchmark cognitive and spatial reasoning, demonstrating that few-shot sampling elevated generative model accuracy by 43.5%. To neutralize the risk of AI "hallucination," cutting-edge frameworks now decouple probabilistic visual perception from deterministic reasoning. Advanced systems utilize the DEXPI (Data Exchange in Process Industry) XML schema to cast geometries and text into semantically enriched Labeled Property Graphs (LPG) via tools like pyDEXPI.

Evolutionary Stage	**Primary Mechanisms**	**Engineering Implementation**	**Failure Modes**
Stage 1: Classical	Otsu's thresholding, Morphological thinning, LBP	Tracing pipe connectivity and identifying contours	Sensitive to scale and rotation;

Evolutionary Stage	Primary Mechanisms	Engineering Implementation	Failure Modes
			destroys flow semantics
Stage 2: Deep Learning	YOLOv3, Faster R-CNN, GANs	High-precision symbol localization	Yields isolated coordinates; lacks functional logic
Stage 3: Topology/OCR	EAST detector, LSTM, Graph modeling	Flow sequences and deep text extraction	Euclidean proximity heuristics fail in dense sections
Stage 4: Graph-RAG	pyDEXPI, Neo4j, LLM, Text2Cypher	Semantic reasoning and zero-hallucination querying	Requires rigid standards-based guardrails

Methodology: The Hybrid Deterministic-RAG Architecture

Strategic industrial deployment requires an architecture that combines the predictive power of probabilistic perception with the absolute reliability of deterministic engineering guardrails. The proposed methodology follows a five-layer processing pipeline that transforms unstructured P&ID pixels into rigorously validated, standards-compliant spreadsheet deliverables.

Layer 1: Perception Layer — Visual and Spatial Extraction

The Perception Layer manages the initial translation of unstructured pixel data into machine-readable primitives. This stage utilizes high-resolution input normalization, strictly maintaining images between 300 and 600 Dots Per Inch (DPI) to preserve microscopic textual annotations.

YOLOv8 Bounding-Box Object Detection

To identify process equipment and instrumentation symbols, the system deploys a fine-tuned YOLOv8 (You Only Look Once, version 8) deep CNN. Unlike earlier anchor-based versions, YOLOv8 utilizes an anchor-free bounding-box regression algorithm operating over a CSPDarknet backbone. This allows the model to globally reason over the diagram simultaneously, identifying diverse symbol classes with extreme speed and scale invariance. The model outputs a defined set of bounding boxes $B = \{x, y, w, h\}$, each defined by spatial coordinates, dimensions, a class label vector (e.g., valve_control), and a probabilistic confidence score.

Advanced OCR Pipeline

Concurrently, a specialized OCR pipeline extracts critical textual annotations. Standard engines often fail on P&IDs due to rotated text physically intersecting with drafting lines. To overcome this, the architecture employs the EAST detector for multi-oriented text localization. Localized regions are then fed into the Tesseract v4 engine, utilizing LSTM sequence recognizers to decode alphanumeric strings. Raw text is passed through Python RegEx pattern matching for post-OCR normalization, correcting common transcription errors (e.g., confusing '1' with 'l') within standard instrument tags.

Layer 2: Topological Layer — LPG Construction and pyDEXPI

Isolated bounding boxes and floating text strings possess no inherent value until structural relationships are mapped. The Topological Layer functions as a multimodal data fusion engine, tasked with assembling disparate artifacts into coherent engineering entities.

Spatial Matching and Semantic Association

To bypass the catastrophic failures of Euclidean distance mapping, this layer utilizes overlap detection via the Python Shapely library. The system calculates a proportional radial boundary for every detected instrument bubble. When an OCR-detected text box spatially intersects or falls within the radius of a specific bubble, a deterministic structural link is formed. This contextual grouping ensures tags are definitively linked to their parent symbols, mitigating the risk of cross-tagging adjacent process lines.

Graph Modeling and Connectivity Reconstruction

Beyond text-to-symbol association, the system reconstructs the physical plant layout using the NetworkX graph library. The P&ID is modeled as a directed graph $G = (V, E)$, where vertices V represent equipment, valves, and instrument nodes, and edges E represent interconnecting pipelines and signal wires. Algorithms trace geometric lines identified during edge detection to establish logical "from-to" fluid flow sequences. This spatial mapping is serialized into a Labeled Property Graph (LPG) and stored in a Neo4j database, enabling complex queries regarding which transmitter sits on which piping segment.

Layer 3: Deterministic Layer — Engineering Rule Validation

The core technological differentiator of this hybrid architecture is the introduction of a non-negotiable Deterministic Validation Layer. Before data is subjected to generative reasoning, it must pass through hard-coded mathematical and lexical guardrails derived directly from global engineering standards.

ISA-5.1 Instrument Tagging Decoding

The system embeds the syntactical rules mandated by ANSI/ISA-5.1-2024 (Instrumentation and Control - Symbols and Identification). These rules are

programmed into a Python-based deterministic parsing algorithm utilizing complex nested RegEx patterns. When the topological layer delivers a fused string (e.g., FIC-101), the engine decomposes the tag into functional constituent parts.

Prefix Letter	Process Variable	Function Letters	Instrument Function	Signal Classification
P	Pressure	T	Transmitter	Analog Input (AI)
F	Flow	IC	Indicator Controller	Analog Input (AI)
L	Level	SHH	Switch High-High	Digital Input (DI)
T	Temperature	CV	Control Valve	Analog Output (AO)
X	Unclassified	V	On/Off Valve	Digital Output (DO)

The algorithm automatically classifies the required field instrument into standardized control system signal typologies: Analog Input (AI), Digital Input (DI), Analog Output (AO), or Digital Output (DO).

IEC 61511 Functional Safety Logic Validation

The IEC 61511 standard rigorously governs how instruments must behave to prevent catastrophic failures. The rule engine algorithmically extracts safety parameters, isolating process safety boundaries based on severity setpoints (High, Low, High-High, Low-Low) identified from visual bubbles on the P&ID. To map multi-variable trip logic, the system cross-references interlocks against logical safety structures, mapping process "Causes" (e.g., PSHH-101 breaching threshold) directly to "Automated Effects" (e.g., immediate closure of Emergency Shutdown Valve SDV-601).

Layer 4: Cognitive Layer — Graph-Based Retrieval-Augmented Generation

To bridge the gap between raw extraction and semantic engineering intent, the architecture leverages advanced generative AI. To neutralize the risk of hallucination, the architecture implements Graph-based Retrieval-Augmented Generation (Graph-RAG), orchestrated via LangChain and LlamaIndex.

Zero-Hallucination Pipeline

The verified topological data and decoded tags in Neo4j serve as immutable factual grounding. When a user query requests a complex deliverable, a Text2Cypher retriever translates the natural language prompt into a Cypher query script. The system retrieves exact, deterministic graph relationships from Neo4j and passes this rigidly constrained

context into the LLM (e.g., GPT-4o). Prompt engineering follows a strict 8-step cognitive reasoning pipeline:

1. **Ingest Knowledge**: Anchors the model to facts retrieved from Neo4j.
2. **Compress**: Filters out irrelevant noise to prevent context window overload.
3. **Extract Facts**: Lists ONLY observable inputs without inferring meaning.
4. **Map to Rules**: Connects facts mathematically to ISA-5.1 deterministic rules.
5. **Apply Constraints**: Restricts the model from guessing missing fields or inventing loops.
6. **Handle Uncertainty**: If evidence is lacking, forces the model to return a REQUIRES_VALIDATION flag.
7. **Verify Output**: Internal self-check for contradictions between logic and evidence.
8. **JSON Formatting**: Returns output as valid JSON, ensuring deterministic lock.

Layer 5: Output Layer — Automated Deliverable Generation

The final stage translates cognitively processed JSON data into deployable engineering formats. Utilizing the Python Pandas library, unstructured JSON arrays are deserialized and cast into structured DataFrames. Backend scripts generate industry-standard Excel and CSV files segregated into specific deliverables:

- **Instrument Index**: Populated with Tag Number, Equipment Description, Fluid Name, and P&ID Reference.
- **I/O List**: Mapping field tags to PLC Module slots and dictating electrical Signal Types (AI, AO, DI, DO) extracted during deterministic validation.
- **Trip List**: Formatting IEC 61511 boolean logic into a standard Cause and Effect matrix.

The Brigade de Cuisine Management Framework

Architectural development follows the specialized "Brigade de Cuisine" system to ensure an organized production line. In this framework, "Recipes" represent the SDLC and engineering standards (ISA/IEC), while "Ingredients" represent raw data quality.

1. **Executive Chef (Lead Architect)**: Defines the RAG pathway and strategic objectives, ensuring models don't hallucinate based on probabilistic weights.
2. **Sous Chef (Standards Lead)**: Translates ISA and IEC manuals into executable Python validation rules, building the deterministic guardrails.

3. **Saucier (Vision Engineer)**: Manages the Perception Layer (YOLO/EAST), refining OCR pipelines for robust text extraction.
4. **Patissier (Data Pipeline Lead)**: Constructs the backend logic for deliverable formatting and presentation.
5. **Expediter (QA Engineer)**: Rotates during final sprint phases to cross-reference AI outputs against ground-truth diagrams to ensure zero-error delivery.

Results and Discussion: Performance Evaluation and Standards Compliance

To evaluate the operational efficacy of the proposed hybrid architecture, a comprehensive prototype was deployed and tested against a robust dataset of high-density industrial P&IDs. Testing assumed stringent benchmarks for spatial localization and semantic validity.

Quantitative Performance of the Perception Layer

The YOLOv8 object detection model demonstrated exceptional stability in identifying graphical geometries, even when subjected to heavy drafting clutter and overlapping annotations. High baseline confidence scores validate the efficacy of the OpenCV preprocessing and anchor-free regression techniques.

Detected Component Class	Mean AI Confidence Score	Bounding-Box Recall Rate
Instrument Bubble (Pressure)	0.95	96.2%
Valve (Control)	0.94	94.8%
Instrument Bubble (Level)	0.91	92.5%
Valve (Check)	0.89	91.1%
Valve (Gate)	0.88	90.4%

Confidence scores—specifically the 0.95 rating for pressure instrument bubbles—indicate that the model successfully generalizes geometric features despite scale variations. Bounding-box recall rates establish the comprehensive spatial baseline necessary for downstream logical validation.

Elimination of False Positives via Topological Fusion

The true success of the prototype is realized in the interaction between Topological, Deterministic, and Cognitive layers. Traditional pipelines relying on Euclidean distance heuristics for text-to-symbol mapping inevitably trigger data integrity failures in dense drawings. A legacy model might erroneously associate FIC-101 with an adjacent pump simply due to physical proximity.

By calculating bounding-box intersections via Shapely and serializing geometry into a Neo4j Labeled Property Graph (LPG), the proposed architecture entirely bypasses this failure mode. When spatial ambiguity occurs, the Graph-RAG cognitive layer queries the explicit piping topology. The LLM reasons over graph edges, recognizing that an instrument bubble must be connected via a capillary line to its respective valve, regardless of raw Euclidean proximity on the sheet. This contextual reasoning reduced text-to-symbol association false positives to a statistically negligible margin.

Zero-Violation Compliance with ISA-5.1 and IEC 61511

Integration of the deterministic rule engine yielded absolute, zero-violation compliance with ANSI/ISA-5.1-2024 tagging conventions. By stripping the LLM of generative freedom and constraining it to hard-coded RegEx libraries, the system mathematically guaranteed that every instrument in the Instrument Index was correctly decoded. Automated classification of signal typologies (AI, AO, DI, DO) was achieved without hallucinations. Because the LLM was restricted by the 8-step pipeline—rejecting uncertainty and flagging ambiguous logic—it never "invented" wiring that did not exist.

Standard	Validation Target	Result	Implication
ISA-5.1	Tag Decoding (Prefix/Function)	100% Accuracy	Correct I/O list generation
ISA-5.1	Signal Classification (AI/DI)	100% Accuracy	Safe BPCS programming
IEC 61511	Cause-Effect Trip Logic	0 Violations	Valid SIF/SIL interlocks
IEC 61511	Alarm Severity (H/HH)	100% Accuracy	Reliable SIS documentation

The system successfully mapped boolean IEC 61511 safety logic, accurately identifying High-High (HH) causes and mapping them to their respective shutdown effects (e.g., closing an SDV) to generate an accurate Cause and Effect matrix without requiring extensive manual cross-checking. The application of Graph-RAG proved transformative in handling off-page connectors spanning multiple P&ID sheets. By executing Cypher queries across Neo4j, the LLM retrieved the full logic loop of a sensor on Sheet 1 triggering a valve on Sheet 5.

Conclusion and Future Scope

This research confirms that a hybrid artificial intelligence architecture can successfully bridge the Contextual Integrity Gap in P&ID digitization. By integrating deep learning perception with deterministic rule engines and Graph-RAG, we have transformed

industrial documentation from a manual labor burden into a reliable digital asset. The manual transcription of P&IDs into functional E&I deliverables represents an archaic, inefficient, and hazardous bottleneck. Purely probabilistic machine vision models fail to capture the overarching control philosophy and safety architecture required for deployment, creating a dangerous integrity gap.

This study successfully engineered and validated a multi-layered hybrid architecture designed specifically to bridge the gap between raw pixel detection and semantic engineering intent. By coupling YOLOv8 and advanced OCR perception with a robust topological modeling engine, the system accurately captures spatial and physical connectivity. Crucially, the deterministic validation layer inherently enforces global industrial standards, mathematically guaranteeing compliance with ISA-5.1 tagging syntax and IEC 61511 functional safety trip logic. The introduction of a Graph-RAG cognitive pipeline operating over a Neo4j database completely neutralizes the risk of generative hallucinations.

The key contributions of this work include the elimination of generative hallucinations through graph-based factual grounding, the automated generation of audit-ready E&I deliverables (Instrument Indexes, I/O Lists) that are 100% compliant with industrial standards, and the implementation of a structured project management framework based on the Brigade de Cuisine system. Future scope for this research will focus on several critical areas:

1. **Perception Robustness**: Training custom YOLO models on more diverse, multi-national drafting datasets to enhance generalization and solve the class imbalance problem through synthetic data generation.
2. **Semantic Expansion**: Expansion of the automation pipeline to include the programmatic generation of complex interlock logic modeling and SIL-level classification.
3. **Interoperability**: Native integration with the DEXPI data model to enhance interoperability across disparate CAD platforms and improve digital twin readiness.
4. **Cybersecurity**: Integration of industrial cybersecurity requirements as outlined in the most recent updates to the IEC 61511 standard and IEC 62443.

As the industry moves toward autonomous engineering, hybrid AI systems will be foundational in ensuring that operational efficiency is never achieved at the cost of functional safety. This study establishes a new benchmark for standards-aligned AI in heavy industry, successfully transitioning legacy documentation into reliable, safety-centric digital assets.